Reg. No. | | | | | | | | | | | [18CS3DS02]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2019

SAMPLING METHODS

Time:3 Hours **Maximum Marks:70**

Instructions:
- **i.** Answer **FIVE** full Questions
- **ii.** Questions 1 and 2 are compulsory.
- **iii.** Missing data may be suitably assumed.

1. a. Define the probability sampling and explain about their types with graphical representations. **07**

b. Define the terms sample size, generalizing, and validity in the context of sampling methods. **07**

2. a. Define the non-probability sampling and explain about their types with graphical representations. **07**

b. Define the terms bias, sampling error, systematic error, and random error. **07**

3. a. Explain the hypergeometric distribution. **08**

A family of eight contains three males and five females. Find the frequency distribution of the number of males in a simple random sample of size 4. Also find the mean and variance for number of males.

b. Formulate the random sampling with replacement in sampling methods. **06**

OR

4 a. Discuss about confidence limits in the perspective of probability distribution. **08**

In a simple random sample of size 100, from a population of size 500, there are 37 units in class *C*. Find the *95%* confidence limits for the proportion and for the total number in class C in the population.

b. Explain about estimation of a ratio in sampling. **06**

5. a. At the end of every school year, the state administers a reading test to a sample of 36 third graders. The school system has 20,000 third graders, half boys and half girls. The results from last year's test are shown in the table below. **09**

Stratum	Mean Score	Standard Deviation
Boys	70	10.27
Girls	80	6.66

This year, the researchers plan to use a stratified sample, with one stratum consisting of boys and the other, girls. Use the results from last year to answer the following questions (Assume a 95% confidence level).

To maximize precision, how many sampled students should be boys and how many should be girls? Find the mean reading achievement level in the population and compute the confidence interval and margin of error.

b. Describe Systematic Sampling and delineate the Steps to form a sample using the Systematic Sampling technique. **05**

OR

6. a. At the end of every school year, the state administers a reading test to a sample of third graders. The school system has 20,000 third graders, half boys and half girls. This year, a proportionate stratified sample was used to select 36 students for testing. Because the population is half boy and half girl, one stratum consisted of 18 boys; the other, 18 girls. Test scores from each sampled student are shown below: **09**

Boys	50, 55, 60, 62, 62, 65, 67, 67, 70, 70, 73, 73, 75, 78, 78, 80, 85, 90
Girls	70, 70, 72, 72, 75, 75, 78, 78, 80, 80, 82, 82, 85, 85, 88, 88, 90, 90

Using sample data, estimate the mean reading achievement level in the population. Find the margin of error and the confidence interval. Assume a 95% confidence level.

b. Explain the cluster sampling with examples and delineate the conditions when to use the cluster sampling. **05**

7. a. Instead of sampling individual passengers, it has been decided that buses (clusters) are going to be sampled and then every passenger on each of the selected buses will be asked if they travelled by bus prior to the project. To calculate the sample size we require the number of clusters that make up the population, that is the number of buses that carry the transport project passengers; for this example we will assume 12,000 buses. Calculate the estimated proportions of passengers that would have travelled by bus prior to the project from a number of buses with 95% confidence level (critical value = 1.96); For this problem we have sampled four buses and the proportions were: **07**

Bus	Estimated Proportion
1	0.37
2	0.46
3	0.28
4	0.52

b. The parameter of interest is the average journey length of people who take local buses. Instead of sampling numerous individual passengers we would like to sample everyone from a few buses (clusters). Knowing that there are 12,000 local buses in Bogota each day, how many buses would we have to sample to find the average journey length with 95% confidence? **07**

The data are available from four buses:

Bus	Total journey length (on average)
1	195
2	96
3	63
4	159

OR

8. a. Consider a scenario. The households are not located in different districts. Instead they are 'clustered' or grouped into lots of villages. Instead of going to numerous individual households, Calculate a number of villages and sample every household within each village with 90% confidence level (critical value = 1.645). The population comprises 120 villages, all of approximately similar size. The proportion of cook stoves still operating and the variation in this proportion between villages are given in the following table: **07**

Village	Estimated proportion of cook stoves operating in each village
1	0.37
2	0.48
3	0.50
4	0.27
5	0.68

b. Suppose CFLs were distributed to households in 50 villages. Instead of sampling from the whole population of households with CFLs, we sample a number of villages (villages=clusters), and then collect data from all households within the villages. How many villages would we have to sample to find the CFL usage at the village level with 90% confidence (critical value = 1.645)? **07**

The data are available from a pilot study on five villages.

Village	Total usage across all households in the village
1	30458
2	27667
3	31500
4	28350
5	19125